# BASIL HAZEM AYYAD

Software Engineer · Backend

Amman, Jordan | leaderbasil@gmail.com | linkedin.com/in/basilhazemayyad | github.com/leaderbasil

## PROFESSIONAL SUMMARY

Software Engineer focused on backend systems and AI agents, with hands on experience building REST APIs, async services, and full stack products across web, mobile, and desktop. Works at Obsidian Labs, a startup in its early stages, building PolarisX ERP from scratch. Also built Arjan, an AI assisted healthcare platform, as a graduation project. Closely follows the global ERP market through hands on demos of leading products.

## PROFESSIONAL EXPERIENCE

**Software Engineer** · Obsidian Labs *Dec 2025 to Nov 2026*

- Built **PolarisX ERP** with the core team, a fully custom enterprise platform from scratch using **Sanic (Python)**, covering inventory, CRM, POS, accounting, operations, reporting, and document management modules.
- Architected scalable REST APIs with Sanic and drove ERP integration with client systems and external data sources.
- Built web and desktop applications with React and Electron, and mobile applications with Flutter, shipping features on startup timelines.
- Led a small mobile team, setting technical direction and reviewing production code.
- Contributed to testing and deployment across the platform.

**Software Engineer Trainee** · 4matex Information Technology *Feb 2026 to Apr 2026*

- Completed an intensive Low Code training program with OutSystems, covering application development, database integration, and UI design.
- Built multiple functional business applications in a condensed training cycle, working with Agile workflows in a team.

## KEY PROJECTS

**PolarisX ERP** · **Lead Backend Engineer**

**Stack:** Sanic, MySQL, SQLAlchemy, REST APIs, JWT Authentication, Flutter, React, Electron

- Designed and built the PolarisX backend from the ground up, an ERP platform for enterprise clients, with React + Electron desktop and Flutter mobile clients.
- Delivered 7 core modules: Inventory, CRM, Accounting, Point of Sale, Operations, Reporting, and Documents.
- Integrated PolarisX with client systems and managed data pipelines between enterprise data sources.
- Website to download PolarisX and see its features: https://polarisx.obsidianlabs.space/

**Arjan · Smart Health Clinic Management System** · **Graduation Project**

**Stack:** FastAPI, PostgreSQL, SQLAlchemy, Gemini API, React, TypeScript

- Designed the backend architecture on FastAPI and PostgreSQL: 56 tables across 10 domains, over 160 endpoints, JWT authentication, role based access control, and audit logging.
- Healthcare platform covering the full patient journey: appointment scheduling, medical records, lab test lifecycle from request to reviewed result, referrals, and clinical decision support (drug interaction checks, order sets).
- Integrated Gemini for AI analysis of lab reports and medical images, alongside a DICOM viewer for radiology imaging.
- Live demo: https://arjan-demo.netlify.app/#/

**RIPPLE** · **Incremental Build Analysis Engine**

**Stack:** C++, Graph Algorithms, Topological Sort, Content Hashing

- Finds exactly which files must rebuild after an edit, and in what order, without launching a compiler, using an include graph, content hashing, and Kahn's algorithm, and reports dependency cycles.
- GitHub: https://github.com/leaderbasil/ripple

## TECHNICAL SKILLS

**Languages:** Python, C++, Go

**Backend:** FastAPI, Sanic, REST APIs, OpenAPI, JWT Authentication, Async Systems, ERP Integration

**Databases:** MySQL, PostgreSQL, SQLite, SQLAlchemy

**Tools:** Git, GitHub, Codeberg, OneDev, Postman, Insomnia, cURL, VS Code

**Practices:** Team Leadership, SDLC, Software Testing, Code Review, Deployment

**Secondary Skills:** React, Electron, Flutter, JavaScript, TypeScript, Dart, OutSystems (Low Code)

**Soft Skills:** Content Strategy, Market & Audience Analysis, Technical Communication, Problem Solving

**Interests:** AI Agents, Backend Architecture, Web Development, ERP Systems, Mobile Development, Continuous Learning

**Looking to learn:** Go, Echo, C#, .NET

## EDUCATION

**Bachelor of Science in Computer Science** *2023 to 2027*

Al al-Bayt University · Grade: Good

## LANGUAGES

Arabic (Native) · English (Professional Working Proficiency)